

Throughout, let $S = k[x_1, \dots, x_r]$ with k algebraically closed, $\mathfrak{m} = (x_1, \dots, x_r)$, let $J \subseteq \mathfrak{m}$ be an ideal, $R = S/J$, $X = \mathbb{V}(J) \subseteq \mathbb{A}^r$, and $I = \mathfrak{m}R$. The goal of this worksheet is to prove

$$\mathrm{gr}_I R \cong S/\mathrm{in}_I(J)$$

and to verify the easy half of the geometric statement on a concrete example.

Problem 1. The Graded Pieces.

- (a) Show that $I^n = (\mathfrak{m}^n + J)/J$ as ideals of $R = S/J$. (Hint: $I = \mathfrak{m}R$ is generated by the images of $x_1, \dots, x_r$; describe I^n as the image of $\mathfrak{m}^n$ under $S \rightarrow R$.)
- (b) Using part (a), establish the isomorphism of k -vector spaces

$$\frac{I^n R}{I^{n+1} R} \cong \frac{\mathfrak{m}^n + J}{\mathfrak{m}^{n+1} + J}.$$

(Hint: apply the third isomorphism theorem to the quotient of quotients.)

- (c) The polynomial ring S is graded by degree, $S = \bigoplus_n S_n$ where S_n is the span of degree- n monomials. Show that $S_n \cong \mathfrak{m}^n / \mathfrak{m}^{n+1}$ as k -vector spaces, and conclude that $\mathrm{gr}_{\mathfrak{m}} S \cong S$ as graded rings.

Problem 2. Initial Forms

- (a) Recall from Section 2 that for $f \in S$ with $f \in \mathfrak{m}^m \setminus \mathfrak{m}^{m+1}$, the initial form $\mathrm{in}(f) \in \mathfrak{m}^m / \mathfrak{m}^{m+1} = S_m$ is the lowest-degree homogeneous part of f . Define a graded ring homomorphism

$$\phi : S \cong \mathrm{gr}_{\mathfrak{m}} S \longrightarrow \mathrm{gr}_I R, \quad S_n \ni \bar{g} \longmapsto (g + \mathfrak{m}^{n+1} + J) \in \frac{I^n R}{I^{n+1} R}.$$

Using Problem 1, show that ϕ is surjective.

- (b) Show that for $g \in S_n$ (homogeneous of degree n),

$$g \in \ker \phi \iff g \in \mathfrak{m}^{n+1} + J$$

Then show that $(\ker(\phi))_n = \mathrm{in}_I(J) \cap S_n$ and thus deduce that $\ker \phi = \mathrm{in}_I(J)$, the ideal generated by $\{\mathrm{in}(f) : f \in J\}$.

- (c) Conclude from parts (a) and (b) that

$$\mathrm{gr}_I R \cong S/\mathrm{in}_I(J).$$

- (d) (*Why all of J is needed.*) Revisit the Remark in Section 2 with $J = (x^2 + y^3, xy) \subset k[x, y]$. Show that $y^4 \in \text{in}_I(J)$ but $y^4 \notin (\text{in}(x^2 + y^3), \text{in}(xy)) = (x^2, xy)$. Explain why this means $\text{gr}_I R$ cannot be computed by only taking initial forms of a chosen generating set of J .

Problem 3.

- (a) Let $f \in J$ with initial form $\text{in}(f) = f_m$ (its degree- m part, $f_m \neq 0$), so $f = f_m + f_{m+1} + \cdots + f_d$ with each f_i homogeneous of degree i . Fix a nonzero direction $p = (p_1, \dots, p_r)$ and consider the points $tp \in \mathbb{A}^r$ for $t \in k$. Compute $f(tp)$ as a polynomial in t and show

$$f(tp) = t^m(f_m(p) + t f_{m+1}(p) + \cdots + t^{d-m} f_d(p)).$$

- (b) Suppose $p_n \in X \setminus \{0\}$ is a sequence with $p_n \rightarrow 0$ and $[p_n] \rightarrow [p]$ in $\mathbb{P}^{r-1}$. Using that $f(p_n) = 0$ for all n (since $p_n \in X$ and $f \in J$), argue informally that $f_m(p) = 0$, i.e. the limiting direction $[p]$ lies in $\mathbb{V}(\text{in}(f))$. Conclude

$$TC_0(X) \subseteq \mathbb{V}(\text{in}_I(J)).$$

(*This is the easy inclusion. The reverse inclusion holds when k is algebraically closed but requires tools beyond these notes.*)

- (c) (*Worked example.*) For $f = y^2 - x^3$, the cuspidal cubic curve, carry out part (a) explicitly: find $\text{in}(f)$, factor it, and identify the limiting secant direction in $\mathbb{P}^1$.